EXACT-DIMENSION OF FURSTENBERG MEASURES

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Let m be a Radon measure on a metric space (X, d).

Definition *The measure m is said to be* **exact-dimensional** *if there is a number δ such that, for m-a.e. $x \in X$,*

$$\delta \;=\; \lim_{r \to 0} \frac{\log m(B(x,r))}{\log r}.$$

The number δ is called the dimension of m. If m is exact-dimensional, all natural notions of m-essential Hausdorff dimension coincide with δ.

Proposition [Young 82] *Let m be exact-dimensional with dimension δ. Then,*

$$\delta \;=\; \inf\{HD(A) : A \subset X,\; A\, Borel,\; m(X \setminus A) = 0\}.$$

Conversely,

Proposition [Frostman's Lemma 35] *Let $K \subset \mathbb{R}^d$ be compact. Then $HD(K)$ is the supremum of the dimensions of the exact-dimensional measures supported on K.*

Comments:

Often, it is easy to estimate from above the Hausdorff dimension of a set, but much harder to estimate it from below. A common trick is to find a suitable exact-dimensional measure.

For random walks, a.e. convergence theorems give exact-dimensionality of stationary measures. In this talk (mostly based on joint work with Pablo Lessa), we discuss several examples.

It is not true, in general, that the HD of an invariant set is given by some stationary measure. This is one of the properties to discuss.

Let G be a connected semi-simple real algebraic group with finite center and no compact factor, μ a probability measure on G with finite support, X a compact metric G-space and ν an extremal stationary probability measure on X: ν satisfies, for all continuous function φ on X

$$\int_X \varphi(\xi)\, d\nu(\xi) \; = \; \sum_G \int_X \varphi(g\xi)\, d\nu(\xi)\; \mu(g)$$

and cannot be decomposed as a convex combination of measures with the same property.

Such measures ν exist by compactness of X. We are interested in the dimension properties of ν.

I. $G = PSL(2, \mathbb{R}), X = \mathbb{P}^1(\mathbb{R})$.

II. $G = SL(3, \mathbb{R}), X = \mathbb{P}^2(\mathbb{R})$.

III. $G = SL(d, \mathbb{R}), d > 3, X = \mathbb{P}^{d-1}(\mathbb{R})$.

IV. General G, X is the flag space G/P. [L-Lessa 24] for $SL(d, \mathbb{R})$, [L-Lessa WIP] for G.

Notations. We consider the Bernoulli measure $m = \mu^{\mathbb{Z}}$ on the sequence space $\Omega = G^{\mathbb{Z}}$ and we let $g_n : \Omega \to G$ be the n-th coordinate projection, T the left-shift transformation on Ω. The *random walk* (G, μ) is the family $Z_n(\omega), n \in \mathbb{Z}, \omega \in \Omega$,

$$Z_n(\omega) := g_{n-1}(\omega) \dots g_0(\omega) \qquad \text{for } n > 0, \quad Z_0 = \mathbb{I}d$$
$$Z_n(\omega) := g_n^{-1}(\omega) \dots g_{-1}^{-1}(\omega) \qquad \text{for } n < 0.$$

For $n, m \in \mathbb{Z}$, we have $Z_{m+n}(\omega) = Z_m(T^n\omega) Z_n(\omega)$.

If there exists a measure dx with full support such that $(Z_n(T^{-n}\omega))_* dx$ $(= (Z_{-n}(\omega)^{-1})_* dx)$ converge to a Dirac measure at $Z_\infty(\omega)$ as $n \to \infty$, then the distribution of $Z_\infty(\omega)$ is a stationary ergodic measure.

Let $H(Y)$ be the Shannon entropy of the finite-valued random variable Y. The *random walk entropy* $h_{RW}(\mu)$ is given by

$$h_{RW}(\mu) := \lim_{n \to +\infty} \frac{1}{n} H(Z_n).$$

If ν is an ergodic stationary measure for (G, μ) on X, the *Furstenberg entropy* of the pair (μ, ν) is defined as

$$\kappa(\mu, \nu) = \sum_{G} \int_{X} \log \frac{dg_* \nu}{d\nu}(g\xi) d\nu(\xi)\, \mu(g),$$

We always have $0 \le \kappa(\mu, \nu) \le h_{RW}(\mu)$ [Derriennic 79, Kaimanovich and Vershik 79].

I. $G = PSL(2,\mathbb{R}), X = \mathbb{P}^1(\mathbb{R})$.

We assume that the group generated by the support of μ is unbounded and totally irreducible. Let λ be the *Lyapunov exponent*: for m-a.e. ω,

$$\lambda \;:=\; \lim_{n\to\infty} \frac{1}{n} \log \|Z_n(\omega)\|.$$

Then, $\lambda > 0$ and $\lim\limits_{n\to\infty} (Z_n(T^{-n}\omega))_* dx$ is the Dirac measure at the unstable direction $[E_1(\omega)]$. The distribution ν of $[E_1(\omega)]$ is the unique stationary measure.

In that setting, ν is exact-dimensional with dimension $\delta \;=\; \frac{\kappa(\mu,\nu)}{2\lambda}$.

Indeed, for most ωs, $Z_n(T^{-n}\omega)$ sends most of the space X on an interval of size $\sim e^{-2\lambda n}$ and ν-measure $\sim e^{-\kappa(\mu,\nu)n}$. But this argument yields only convergence in measure, not ν-a.e. (cf.[L83]).

The complete argument follows [L.-Young 85] for C^2-diffeos, [Feng-Hu 09] for IFS and is done in ([Hochman-Solomyak 17, Section 3.3]).

Ingredients for the proof of $\inf_{\nu\text{-a.e.}x} \liminf_{r\to 0} \frac{\log \nu(B(x,r))}{\log r} \geq \frac{\kappa(\mu,\nu)}{2\lambda}$:

1. There exists a sequence $q_n(\omega)$, with $\lim_{n\to+\infty} \frac{1}{n}\log q_n(\omega) = 2\lambda$, such that

$$\begin{aligned} \nu(B(E_1(\omega), q_n)) &= \Pi_{j=0}^{n-1} \frac{\nu(B(E_1(T^j\omega), q_{n-j}))}{\nu(B(E_1(T^{j+1}\omega), q_{n-j-1}))} \\ &\leq \Pi_{j=0}^{n-1} \frac{\nu(B(E_1(T^j\omega), q_{n-j}))}{(g_j^{-1})_*\nu(B(E_1(T^j\omega), q_{n-j}))} \end{aligned}$$

2. The weak (1,1) convergence in the Lebesgue density theorem: if $\varepsilon_n \to 0$,, $\varphi_{\varepsilon_n}(\omega,\xi) := -\log \frac{(g^{-1})_*\nu(B(\xi,\varepsilon_n))}{\nu(B(\xi,\varepsilon_n))}$, then $\lim_{n\to\infty} \varphi_{\varepsilon_n} = -\log \frac{d(g^{-1})_*\nu}{d\nu}$ and $\sup_n \varphi_{\varepsilon_n} \in L^1(m\times\nu)$.

3. The Maker ergodic theorem applied to $\{\varphi_{q_{n-j}}\}$ on $(\Omega\times X, m\times\nu)$

$$\lim_{n\to\infty} \frac{1}{n}\sum_{j=0}^{n-1} \varphi_{q_{n-j}}(T^j\omega, Z_j(\omega)\xi) = -\int_{\Omega\times X} \log \frac{d(g_0^{-1})_*\nu}{d\nu}\, d\nu(\xi)dm(\omega) \ = \kappa(\mu,\nu).$$

Actually [H-S 17] proves a much more precise result when μ is *Diophantine*

Definition *The measure* μ *is called Diophantine if there exists* $c > 0$ *such that for all* n, $\mu \otimes \mu$*-a.e.* (ω, ω') *with* $Z_n(\omega) \neq Z_n(\omega')$,

$$\|Z_n(\omega) - Z_n(\omega')\| \; > \; c^n.$$

Theorem [Hochman-Solomyak 17] *Assume moreover that the measure* μ *is Diophantine. Then the dimension of* ν *is given by*

$$\delta \; = \; \min\{1, \frac{h_{RW}(\mu)}{2\lambda}\}.$$

II. $G = SL(3, \mathbb{R}), X = \mathbb{P}^2(\mathbb{R})$.

Consider the m-a.e. Lyapunov exponents defined by

$$\begin{aligned}
\lambda_1 &:= \lim_{n\to+\infty} \frac{1}{n} \log \|Z_n(\omega)\| \\
\lambda_1 + \lambda_2 &:= \lim_{n\to+\infty} \frac{1}{n} \log \| \wedge^2 Z_n(\omega)\| \\
\lambda_1 + \lambda_2 + \lambda_3 &= 0.
\end{aligned}$$

Assume $\lambda_1 > \lambda_2 > \lambda_3$ and let $E_1(\omega), E_2(\omega), E_3(\omega)$ be the (random) Oseledets directions: for $j = 1, 2, 3$, for m-a.e. ω,

$$E_j(\omega) \setminus \{0\} \;=\; \{v \in \mathbb{R}^3 : \lim_{n\to\pm\infty} \frac{1}{n} \log |Z_n(\omega)v| = \lambda_j\}.$$

Remark 1. For most ωs, $Z_n(T^{-n}\omega)$ now sends most of the space X on a rectangle of ν-measure $\sim e^{-\kappa(\mu,\nu)n}$ and of sides of sizes $\sim e^{(\lambda_3-\lambda_1)n}$ and $\sim e^{(\lambda_2-\lambda_1)n}$. The one-dimensional arguments cannot directly work.
2. The distribution ν of the direction $[E_1(\omega)]$ is a stationary measure.

Theorem [Feng 23 for IFS, Rapaport 21] *Assume* μ *is totally irreducible. Then the measure* ν *is exact-dimensional with dimension* δ*. There are numbers* $\gamma_1, \gamma_2, 0 \leq \gamma_1, \gamma_2 \leq 1$*, such that*

$$\delta \;=\; \gamma_1 + \gamma_2, \quad \kappa(\mu, \nu) \;=\; (\lambda_1 - \lambda_3)\gamma_1 + (\lambda_1 - \lambda_2)\gamma_2.$$

The full content of the theorem is the geometric meaning of the γ_i.

Basic idea(After [Feng 23, Rapaport 21, L-Lessa 23])

There is a **random one-dimensional foliation** *of* ν*-almost all* $\mathbb{P}^2$ *with dimension conservation.*

Dimension conservation ([Furstenberg])

Definition *Assume that* $(X, d), (X', d')$ *are separable metric spaces and that* $\pi : (X, d) \to (X', d')$ *is a Lipschitz mapping. Let* ν *be a probability measure on* X*. We say that the projection* π *is* **dimension conserving** *for* ν *if*

1. the measure ν is exact-dimensional with dimension δ,
2. the measure $\pi_*\nu$ is exact-dimensional with dimension δ',
3. for $\pi_*\nu$-a.e. $x' \in X'$, the disintegration $\nu^{x'}$ is exact-dimensional on $\pi^{-1}(x')$ with dimension $\delta - \delta'$.

Dimension conservation does not hold in general.

Example: Typical graph of Brownian motion
$B(\omega, t), 0 \leq t \leq 1, B(\omega, 0) = 0$, and $m(\omega)$ lifted from Lebesgue on $[0, 1]$.

For a.e. ω, $m(\omega)$ is exact-dimensional with dimension $3/2$ (Taylor 53).

For the vertical projection, there is no dimension conservation: $\delta' = 1$ and the conditional measures are Dirac, with dimension 0.

For the horizontal projection, there is dimension conservation: almost all Local Times at level y have dimension $1/2$ (Taylor 55)

Proof of the theorem. Recall that ν is the distribution of $E_1(\omega)$ and that $E_3(\omega)$ is independent of $E_1(\omega)$. By total irreducibility, ν is continuous and, $\nu([E_3(\omega)]) = 0$. For $\omega_+ \in G^{\mathbb{N}}$, let $\pi(\omega_+)$ be the projection from $\mathbb{P}^2(\mathbb{R}) \setminus [E_3(\omega_+)]$ to the set of projective planes going through $E_3(\omega_+)$.

Then, $\pi(T\omega_+)$ is conjugated to $\pi(\omega_+)$ by g_0. Moreover, the projected measures $\nu_{\omega+}$ and the conditional measures $\nu_{\omega+}^y$ are equivariant. They defined one-dimensional actions with asymptotic contracting coefficients $(\lambda_2 - \lambda_1)$ and $(\lambda_3 - \lambda_1)$. By the previous one-dimensional argument, a.e. projected measures $\nu_{\omega+}$ and conditional measures $\nu_{\omega+}^y$ are exact dimensional. Moreover, the dimensions γ_2 and γ_1 are a.e. constant and are related to entropies κ_2 and h_1.

$$\gamma_2(\lambda_1-\lambda_2) \quad = \quad \kappa_2 := \int_\Omega \left(\int_{\pi(\omega^+)(X)} \log \frac{d(g_0)_*\nu_{\omega^+}}{d\nu_{T\omega^+}}(g_0x')d\nu_{\omega^+}(x') \right) dm(\omega)$$

$$\gamma_1(\lambda_1-\lambda_3) \quad = \quad h_1 := \int \left(\int \log \frac{d(g_0)_*\nu^{x'}_{\omega^+}}{d\nu^{g_0x'}_{T\omega^+}}(g_0\xi)d\nu^{x'}_{\omega^+}(\xi) \right) dm(\omega)d\nu_{\omega^+}(x').$$

Remains to show that for a.e. ω_+, $\pi(\omega_+)$ is dimension conserving,

1. From the above formulas, $\kappa(\mu,\nu) = h_1+\kappa_2$.
2. By general nonsense, $\underline{\delta} := \inf_{\nu\text{-a.e.}x} \liminf_{r\to 0} \frac{\log \nu(B(x,r))}{\log r} \geq \gamma_1+\gamma_2$.
3. By a counting argument, for $\overline{\delta} := \sup_{\nu\text{-a.e.}x} \limsup_{r\to 0} \frac{\log \nu(B(x,r))}{\log r}$, we have:

$$\overline{\delta}(\lambda_1-\lambda_2) \leq \gamma_1(\lambda_1-\lambda_2)-h_1+\kappa = (\gamma_1+\gamma_2)(\lambda_1-\lambda_2).$$

For the counting argument, we want to estimate from below
$\nu(B(x,r)) = m(\{\omega' : d(E_1(\omega'), E_1(\omega) < r\})$ for m-a.e. ω
$= m(\{\omega' : d(Z_n E_1(T^{-n}\omega'), E_1(\omega) < r\})$ for $r \sim e^{-n(\lambda_1-\lambda_2)}$.

The contraction $(\lambda_3 - \lambda_1)$ is stronger in the fibers than in the quotient space. Thus, for most ω' with $(\omega')^+ = \omega^+$ and $d(Z_n E_1(T^{-n}\omega'), E_1(\omega) < r\})$, $d_1(Z_n E_1(T^{-n}\omega'), E_1(\omega') < 3r$.
For most fibers y, the measure $\nu^y_{\omega^+}$ of that ball of radius $3e^{-n(\lambda_1-\lambda_2)}$ is $\sim e^{-n\gamma_1(\lambda_1-\lambda_2)}$.
There is at least one good fiber y such that, for most cylinders $C_n(\omega') := [g_{-n}(\omega'), \dots g_{-1}(\omega')]$, the conditional measure $m[C_n(\omega') | \pi(\omega^+)(E_1(\omega')) = y]$ is $\sim e^{-nh'_1}$ and the m-measure is $\sim e^{-nh_{RW}(\mu)}$. So there are at least $\sim e^{-n\gamma_1(\lambda_1-\lambda_2)+nh'_1}$ such cylinders and the total ν-measure is indeed at least $e^{-n\gamma_1(\lambda_1-\lambda_2)+nh'_1-nh_{RW}}$.

Finally, from general entropy theory, $h_{RW} - h'_1 = \kappa - h_1$. $\square$

The formula is more precise when the measure μ is Diophantine.

Theorem. [Li, Pan, Xu 24] *Assume that the support of μ is Diophantine in $SL(3,\mathbb{R})$ and that it generates a Zariski dense semi-group Γ. Then the dimension δ of the unique stationary measure ν is given by:*

$$\delta = \begin{cases} \frac{\kappa(\mu,\nu)}{\lambda_1-\lambda_2} & \text{if} \quad \kappa(\mu,\nu) \leq (\lambda_1-\lambda_2) \\ 1+\frac{\kappa(\mu,\nu)-(\lambda_1-\lambda_2)}{\lambda_1-\lambda_3} & \text{if} \quad \kappa(\mu,\nu) \geq (\lambda_1-\lambda_2) \end{cases}.$$

Comments

1. The expression for δ is the largest possible value of $\gamma_1+\gamma_2$ with $0 \leq \gamma_1, \gamma_2 \leq 1$ and $\kappa(\mu,\nu) = \gamma_1(\lambda_1-\lambda_3)+\gamma_2(\lambda_1-\lambda_3)$.

2. The analog result for 2-dimensional affine IFS was due to [Bárány, Hochman, Rapaport 19].

3. This implies ([Jiao, Li, Pan, Xu 24]) a formula for the Hausdorff dimension of the limit set if the group Γ is Anosov.The Hausdorff dimension of the Rauzy gasket (cf [Jurga 23]) can also be obtained as a corollary.

III. $G = PSL(d, \mathbb{R}), d > 3, X = \mathbb{P}^{d-1}(\mathbb{R})$.

Let μ be a probability measure on $PSL(d, \mathbb{R})$with finite support. Let $\lambda_1 \geq \ldots \geq \lambda_d$ be the Lyapunov exponents: for m-a.e. ω, $1 \leq j \leq d$,

$$\lambda_1 + \ldots + \lambda_j := \lim_{n \to +\infty} \frac{1}{n} \log \| \wedge^j Z_n(\omega) \|,$$

$\rho_1 > \ldots > \rho_k$ the distinct Lyapunov exponents, each with multiplicity d_j. Oseledets theorem yields a random a.e. decomposition of $\mathbb{R}^d = E_1(\omega) \oplus \ldots \oplus E_k(\omega)$ with $\dim E_j = d_j$ and, for m-a.e. ω,

$$E_j(\omega) \setminus \{0\} \ = \ \{v \in \mathbb{R}^3 : \lim_{n \to \pm\infty} \frac{1}{n} \log |Z_n(\omega) v| = \rho_j \}.$$

Theorem [F., R., L.-L.] *Assume $d_1 = 1$ and let ν be the distribution of $[E_1(\omega)]$ on $\mathbb{P}^{d-1}(\mathbb{R})$. Then, the measure ν is exact-dimensional with dimension δ. There are numbers $\gamma_1, \ldots, \gamma_{k-1}$, $0 \leq \gamma_j \leq d_j$, such that*

$$\delta \ = \ \sum_{j=1}^{k-1} \gamma_j, \quad \kappa(\mu, \nu) \ = \ \sum_{j=1}^{k-1} (\rho_1 - \rho_{k-j+1}) \gamma_j.$$

For the proof, we construct increasing random foliations $\mathcal{W}^j(\omega^+)$ with dimension conservation such that γ_j are the associated transversal dimensions.

Proof. Let $f'(\omega^+) = \{0\} \subset U'_1(\omega^+) \subset \ldots \subset U'_{k-1}(\omega^+) \subset \mathbb{R}^d$, be the stable flag, where $U'_j(\omega^+) := [\oplus_{i=k-j+1}^k E_i(\omega^+)]$.

Recall that ν is the distribution of $[E_1(\omega)]$ and that $f'(\omega^+)$ is independent of $[E_1(\omega)]$. By total irreducibility, $\nu([U'_{k-1}]) = 0$ and, for $\omega^+ \in G^{\mathbb{N}}$, let $\pi^j(\omega^+)$ be the projection from $\mathbb{P}^{d-1}(\mathbb{R}) \setminus [U'_j(\omega^+)]$ to the set of $(1 + \sum_{\ell=k+j-1}^k d_\ell)$-dimensional spaces containing $U'_j(\omega^+)$. Then, $\pi^j(T\omega^+)$ is conjugated to $\pi^j(\omega^+)$ by g_0.

The fibers of the projection $\pi^j(\omega^+)$ form the random foliation $\mathcal{W}_j(\omega^+)$ of $\mathbb{P}^{d-1}(\mathbb{R}) \setminus [U'_j(\omega^+)]$ by

$$[x] \sim^{\mathcal{W}_j} [x'] \iff \dim(x + x' + U'_j(\omega^+)) = 1 + \dim U'_j(\omega^+).$$

Consider for a.e. y_{k-j}, the conditional measures $\nu_{\omega+}^{j,y_{k-j}}$ on $(\pi^j)^{-1}(y_{k-j})$ and the projected measures $\nu_{\omega+}^j$ of $\nu_{\omega+}^{j,y_{k-j}}$ transversally to $\mathcal{W}^{j-1}$. We can define the conditional entropy h_j and the transverse entropy κ_{j+1} by the same formulas. We still have

$$h_{j+1} \ = \ h_j + \kappa_{j+1}, \quad \kappa(\mu, \nu) = h_1 + \sum_{j=1}^{k-2} \kappa_{j+1}.$$

The measures $\nu_{\omega^+}^{1,y_{k-1}}$ are equivariant, with asymptotic contraction $(\rho_k - \rho_1)$. We show as before that, for m-a.e. ω, the measures $\nu_{\omega^+}^{1,y_{k-1}}$ are exact-dimensional with dimension γ_1 and that $(\rho_1 - \rho_k)\gamma_1 = h_1$.

Assume inductively that, for $\ell \le j$, for m-a.e. ω, the measures $\nu_{\omega^+}^{\ell,y_{k-\ell}}$ are exact-dimensional with dimension $\delta_\ell := \gamma_1 + \cdots + \gamma_\ell$ and that $(\rho_1 - \rho_{k-\ell+1})\gamma_\ell = \kappa_\ell$.

The proof that, for m-a.e. ω, the measures $\nu_{\omega^+}^{j+1,y_{k-j-1}}$ are exact-dimensional with dimension $\delta_j + \gamma_{j+1}$ and that $(\rho_1 - \rho_{k-j+1})\gamma_{j+1} = \kappa_{j+1}$ is parallel to the proof in dimension 3.

1. By the induction hypothesis, for m-a.e. ω, the measures $\nu_{\omega^+}^{j}$ are exact-dimensional with dimension δ_j.
2. The transversal action has a unique exponent $(\rho_{k-j+1} - \rho_1)$ and the key one-dimensional argument yields $(\rho_1 - \rho_{k-j+1})\gamma_{j+1} = \kappa_{j+1}$
3. $\underline{\delta}_{j+1} \ge \delta_j + \gamma_{j+1}$.
4. By the same counting argument, $\overline{\delta}_{j+1}(\rho_1 - \rho_{k-j+1}) \le \delta_j(\rho_1 - \rho_{k-j+1}) - h_j + h_{j+1}$.

The counting argument works since the fiber contraction is stronger than the transversal contraction.

Finally, comparing the above relations proves the inductive step. So, letting $j = k$, for m-a.e. ω, the measures $\nu^k_{\omega+}$ are exact-dimensional with dimension $\delta_k := \sum_{j=1}^k \gamma_j$. By independence, $\nu^k_{\omega+} = \nu$ for m-a.e. ω_+.

Moreover,
$\kappa(\mu, \nu) \; = \; h_k \; = \; h_1 + \sum_{j=1}^{k-2} \kappa_{j+1} \; = \; \sum_{j=1}^{k-1} (\rho_1 - \rho_{k-j+1})\gamma_j$. $\square$

Remark 3 For m-a.e. ω, the foliations $\mathcal{W}^j(\omega^+)$ have a global dynamical description: for $x_1, x_2 \in \mathbb{P}^{d-1}(\mathbb{R}) \setminus [U'_j(\omega^+)]$,

$$x_1 \sim^{\mathcal{W}^j(\omega^+)} x_2 \iff \limsup_{n\to\infty} \frac{1}{n} d(Z_n(\omega)x_1, Z_n(\omega)x_2) \; \leq \; \rho_{k-j+1} - \rho_1 .$$

Remark 4 For all $f' \in \mathcal{F}$, all j, the foliation $\mathcal{W}^j$ is an algebraic foliation of $\mathcal{F} \setminus \{f'\}$ (explicit polynomial coordinates are given in [LL24]).

IV. General G, X is the flag space G/P. [L-Lessa 24] for $SL(d,\mathbb{R})$, [L-Lessa WIP] for G.

Notations. Let G be a connected algebraic semi-simple real algebraic group with finite center and no compact factor and let $\mathfrak{g}$ be its Lie algebra,
K a maximal compact subgroup with Lie algebra $\mathfrak{k}$,
$\mathfrak{a}$ a *Cartan subspace,* a maximal commutative subalgebra of $\mathfrak{g}$ orthogonal to $\mathfrak{k}$.
Σ the set of *restricted roots* and Σ^+ a chosen subset of positive roots. Consider the corresponding Weyl chamber $A^+ := \exp\,\mathfrak{a}^+$, where

$$\mathfrak{a}^+ \;:=\; \{a \in \mathfrak{a} : \alpha(a) \geq 0\,\forall \alpha \in \Sigma^+\}$$

and let Π be the set of positive simple roots (i.e. that cannot be written as the sum of two positive roots.)
Cartan decomposition For $g \in G$, there exists a unique $\gamma(g) \in \mathfrak{a}^+$ such that g can be written $g = k_1 \exp\gamma(g)\, k_2$, with $k_1, k_2 \in K$.
The *Weyl group* $W := \mathrm{Norm}_K(A)/\mathrm{Cent}_K(A)$ preserve A. By definition, $\mathrm{Ad}(W)$ preserves $\mathfrak{a}$. There is a unique element $w_0 \in W$ such that $\iota := -\operatorname{Ad} w_0$ preserves $\mathfrak{a}^+$.
The (dual) mapping ι preserves Σ^+ and Π. If $g = k_1 \exp\gamma\, k_2$ is a KA^+K decomposition of g, then a KA^+K decomposition of g^{-1} is given by $g^{-1} = k_1' \exp(\iota(\gamma))\, k_2'$.

Notations (continued).

For a restricted root $\alpha \in \Sigma$, let $\mathfrak{g}^\alpha$ be the root space associated to α. We write

$$\mathfrak{p} := \mathfrak{m} \oplus \mathfrak{a} \oplus_{\alpha\in\Sigma^+} \mathfrak{g}^\alpha, \quad \mathfrak{p}^- := \mathfrak{m} \oplus \mathfrak{a} \oplus_{\alpha\in\Sigma^-} \mathfrak{g}^\alpha, \quad \mathfrak{n}^\pm := \oplus_{\alpha\in\Sigma^\pm} \mathfrak{g}^\alpha,$$

where $\mathfrak{m} := \mathrm{Cent}_{\mathfrak{k}}(\mathfrak{a})$ and $\Sigma^- := -\Sigma^+$. We also write N^+, N^- for the corresponding connected subgroups of G, M for the commutator of A in K, $P := MAN^+, P^- = MAN^-$. The group P (resp. P^-) is the normalizer of $\mathfrak{p}$ (resp. $\mathfrak{p}^-$) for the adjoint action of G.
Iwasawa decomposition There is a homeomorphism from $K \times \exp(\mathfrak{a}) \times N^+$ (resp. $K \times \exp(\mathfrak{a}) \times N^-$) to G given by the product in G.
Let $\mathcal{F} := G/P$ be the *flag variety* of G, f_P the identity coset in $\mathcal{F}$. Observe tha, since $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{p}$, the tangent space $T_{f_P}\mathcal{F}$ is identified with $\mathfrak{n}^-$.

For any $g \in G$ and $f \in \mathcal{F}$, there is a unique element $\sigma(g, f) \in \mathfrak{a}$ such that, if $f = kf_P$ for some $k \in K$, then $gk \in K \exp(\sigma(g, f))N^+$. The mapping $(g, f) \mapsto \sigma(g, f)$ satisfies the *cocycle relation*:

$$\sigma(gg', f) = \sigma(g, g'f) + \sigma(g', f) \tag{1}$$

(see e.g. [Benoist-Quint 16], Lemma 6.29). It is called the *Iwasawa cocycle.*

Let μ be a probability measure on G with finite support. Assume that the support of μ generates a Zariski dense subgroup of G. Then,

Proposition. *There exists $\lambda \in \mathfrak{a}^{++}$ such that for m-a.e. ω,*

$$\lambda = \lim_{n \to +\infty} \frac{1}{n} \gamma(Z_n(\omega)), \quad \iota(\lambda) = \lim_{n \to +\infty} \frac{1}{n} \gamma(Z_{-n}(\omega)).$$

Existence of a limit in $\lambda \in \mathfrak{a}^+$ follows from the subadditive ergodic theorem; it belongs to the interior $\mathfrak{a}^{++}$ of $\mathfrak{a}^+$ by Zariski density ([BQ16, Theorem 10.9]).

The main argument for an Oseledets theorem in that setting goes back to [Kaimanovich 87]. A complete statement was given by [Alves and San Martin 13]. We use a partial form of it.

Let $\mathcal{F}^{(2)}$ be the only open orbit of the action of G on $\mathcal{F} \times \mathcal{F}$. We say that f and f' are *general position* if $(f, f') \in \mathcal{F}^{(2)}$. For $f \in \mathcal{F}$, we write C_f for the set of flags in general position with f.

Theorem *With the above notations, for m-a.e. ω, there is an unstable flag $f(\omega)$ and a stable flag $f'(\omega)$ such that*

$$f \in C_{f'(\omega)} \iff \lim_{n \to +\infty} \frac{1}{n} \sigma(Z_n(\omega), f) = \lambda$$

$$f' \in C_{f(\omega)} \iff \lim_{n \to +\infty} \frac{1}{n} \sigma(Z_{-n}(\omega), f') = \iota(\lambda).$$

Moreover, $f(\omega) = \lim_{n\to+\infty} (Z_n(T^{-n}\omega))_* dx$ *depends only on the negative coordinates* $\{g_n\}_{n<0}$, $f'(\omega) = \lim_{n\to+\infty} (Z_n^{-1}(T^n\omega))_* dx$ *depends only on* ω^+ *and* $f(\omega)$ *and* $f'(\omega)$ *are in general position. Thus, for* m*-a.e.* ω, *we have*

$$\lim_{n\to+\infty} \frac{1}{n}\sigma(Z_n(\omega), f(\omega)) \;=\; \lambda \textit{ and } \lim_{n\to+\infty} \frac{1}{n}\sigma(Z_{-n}(\omega), f'(\omega)) = \iota(\lambda).$$

Let ν be the distribution in $\mathcal{F}$ of the unstable flag $f(\omega)$. Then

Corollary *We have* $\lambda = \sum_G \left(\int_{\mathcal{F}} \sigma(g,f)\, d\nu(f)\right) \mu(g)$.

Theorem[L.-Lessa] *The measure* ν *is exact-dimensional with dimension* δ. *Let* $\chi_1 < \ldots < \chi_j < \ldots < \chi_k < 0$ *the distinct values of* $\alpha(\lambda), \alpha \in \Sigma^-$. *There are numbers* $\gamma_1, \ldots, \gamma_k$ *such that*

$$\delta \;=\; \sum_{j=1}^{k} \gamma_j, \quad \kappa(\mu,\nu) \;=\; -\sum_{j=1}^{k} \chi_j \gamma_j$$

and, for m*-a.e.* ω^+, *an increasing family of random foliations* $\mathcal{W}^j(\omega^+)$ *with dimension conservation such that* γ_j *is the exact-dimension of cond. measures on* $\mathcal{W}^j(\omega^+)$ *transversally to* $\mathcal{W}^{j-1}(\omega^+)$.

By convention, $\mathcal{W}^0(\omega^+) := \{C_{f'(\omega^+)}\}$, $\mathcal{W}^k(\omega^+)$ is the trivial foliation of $C_{f'(\omega^+)}$ into points.

Proof. There is a real vector space $\widetilde{V}$ and a representation $\Phi : G \to SL(\widetilde{V})$ such that there is an algebraic, equivariant, one-to-one mapping $\eta : \mathcal{F} = G/P \to \mathbb{P}(\widetilde{V})$ and the group generated by the support of $\widetilde{\mu} := \Phi_\star \mu$ is proximal.

(Take $\widetilde{V} := \otimes_{\alpha \in \Pi}(V_\alpha, \tau_\alpha)$, where (V_α, τ_α) is the Tits representation in [BQ16].)

We put on $\widetilde{V}$ an Euclidean metric invariant under $\Phi(K)$ and choose $\widetilde{A}$ and $\widetilde{A}^+$ compatible with $\Phi(A)$ and $\Phi(A^+)$. Then, the application η is bi-Lipschitz and moreover, for m-a.e. ω, $\eta(f(\omega))$ is the one-dimensional direction $[\widetilde{E}_1(\omega)]$ of the highest exponent $\widetilde{\lambda}_1$ of the random walk $(SL(\widetilde{V}), \widetilde{\mu})$.

It follows from Section **III** that $\nu = (\eta^{-1})_* \widetilde{\nu}$ is exact-dimensional and that there are dimension conserving associated non-decreasing foliations $\overline{\mathcal{W}}^p := \eta^{-1}(\widetilde{\mathcal{W}}^p)$ with transversal contractions $\widetilde{\lambda}_{\ell-p+1} - \widetilde{\lambda}_1, 1 < p \leq \ell$.

The claim in the theorem is that the $\overline{\mathcal{W}}^p, 1 \leq p \leq \ell$, are in fact indexed by $1 \leq j \leq k$ and that only the χ_j appear as transversal contractions.

By remark 3, for $1 \leq p \leq \ell$, for m-a.e. ω, the foliations $\overline{\mathcal{W}}^p(\omega^+)$ have a global dynamical description: for $x_1, x_2 \in \mathcal{F} \setminus f'(\omega^+)$,

$$x_1 \sim^{\overline{\mathcal{W}}^p(\omega^+)} x_2 \iff \limsup_{n\to\infty} \frac{1}{n} d(Z_n(\omega)x_1, Z_n(\omega)x_2) \ \leq \ \widetilde{\lambda}_{\ell-p+1} - \widetilde{\lambda}_1.$$

In particular, the leaves $\overline{W}^p$ are connected and simply connected. Moreover, by remark 4, the foliations $\overline{\mathcal{W}}^p(\omega^+)$ are algebraic. So, on $\mathcal{F} \setminus f'(\omega^+)$, the leaf $\overline{W}^p(x)$ is determined by the tangent subspaces $T_x\overline{W}^p \subset T_x\mathcal{F}$. Moreover, we have, for m-a.e. ω, for $x \in \mathcal{F} \setminus f'(\omega^+)$,

$$T_x\overline{W}^p(\omega^+) \ = \ \{v \in T_x\mathcal{F} : \ \limsup_{n\to\infty} \frac{1}{n}|D_x Z_n(\omega)v| \ \leq \ \widetilde{\lambda}_{\ell-p+1} - \widetilde{\lambda}_1\}. \quad (2)$$

On the other hand, recall that $T_x C_{f'(\omega+)}$ is identified with $\mathfrak{n}^-$ by left-invariance. For $1 \leq j \leq k$, let

$$S_j := \{\alpha : \alpha \in \Sigma^-, \alpha(\lambda) \leq \chi_j\} \quad \text{and} \quad \mathfrak{n}_j := \sum_{\alpha \in S_j} \mathfrak{g}^\alpha.$$

Then, $\mathfrak{n}_j$ are sub-Lie algebras of $\mathfrak{n}^-$ and the orbits of $N_j = \exp \mathfrak{n}_j$ on $C_{f'(\omega^+)}$ define an increasing family $\mathcal{W}^j(\omega^+), 1 \leq j \leq k$. Using Oseledets theorem, one sees that

$$T_x W^j(\omega^+) \;=\; \{v \in T_x\mathcal{F} : \; \limsup_{n\to\infty} \frac{1}{n} |D_x Z_n(\omega) v| \;\leq\; \chi_j\}. \tag{3}$$

The conclusion follows from comparing (2) and (3). $\square$

Lyapunov dimension

Definition *Let* (G, μ, λ) *be as above,* $\chi_1 < \ldots < \chi_j < \ldots < \chi_k < 0$ *the distinct values of* $\alpha(\lambda), \alpha \in \Sigma^-$, *with multiplicity* $d_j, 1 \leq j \leq k$.
The **Lyapunov dimension** *of* μ *is the largest value that can take the dimension* $\delta = \sum_{j=1}^{k} \gamma_j$ *for numbers* $\gamma_j, 1 \leq j \leq k$ *such that*

$$0 \leq \gamma_j \leq d_j, \quad \kappa(\mu, \nu) = -\sum_{j=1}^{k} \chi_j \gamma_j .$$

Conjecture. *Assume that the support of* μ *is Diophantine in* G *and that it generates a Zariski dense semi-group* Γ. *Then the dimension* δ *of the unique stationary measure* ν *is given by the Lyapunov dimension of* μ.

In particular, for Anosov representations, the conjecture would lead to a formula à la Sullivan for the Hausdorff dimension of the limit set, subsuming results by B Pozzetti, A. Sambarino, and A. Wienhard ([PSW23]), O. Glorieux, D. Monclair, and N. Tholozan ([GMT23]), Y. Jiao, J. Li, W. Pan, and D. Xu ([JLPX24]) and G. Stamatiou ([S26]).